Redger Xu

redgerxu@berkeley.edu | github.com/regarager | (408) 769-0763

EDUCATION

University of California, Berkeley | B.S. Electrical Engineering & Computer Sciences — Class of 2030

- Current Coursework: Data Structures, Linear Algebra & Differential Equations

Stratford Preparatory Blackford | High School Diploma (GPA: 4.87) — June 2026

- Relevant Coursework: AP Computer Science A, AP Calculus BC, AP Statistics, Cybersecurity, Full-Stack Development, AI: Models and Applications, Data Structures and Algorithms

PROJECTS

ClashAI (code) *(Reinforcement Learning + Computer Vision)* — Fall 2025 – Spring 2026

- Investigated real-time perception–action bottlenecks in game-playing RL agents; benchmarked ADB versus minicap screen-capture frameworks, reducing end-to-end latency from ≈ 200 ms to ≈ 30 ms while preserving YOLOv11 detection integrity
- Implemented PPO with multi-head attention and action masking for legal move selection; engineered jitter-robust card recognition via dHash matching with Sobel edge detection and Otsu thresholding
- Conducted ablation studies on latency-vs-accuracy trade-offs, identifying the capture pipeline as the primary bottleneck for real-time decision loops

MaLO | Contributor *(LLMs in Cybersecurity)* — Summer 2025

- Integrated the OpenRouter API to enable benchmarking across multiple LLM models; controlled system prompt, temperature, and other key hyperparameters across models
- Enforced reproducibility across different models using Docker and standardized challenge files
- Created pipeline to run all tests automatically, removing manual per-run intervention and speeding up tests

Analysis of California Traffic Collisions *(Data Science + Forecasting)* — Summer 2024

- Engineered multi-class classification and time-series forecasting pipelines on 5M+ California Highway Patrol collision records to identify environmental, temporal, and vehicular risk factors
- Designed a custom "lenient accuracy" metric to handle near-correct classifications; benchmarked Logistic Regression, SVM, and Random Forest
- Built a time-series forecasting model using skforecast with lag optimization; formulated a causal hypothesis on Daylight Saving Time (DST) shifts, classifying post-DST periods vs. normal days with 71.4% accuracy

IntBee (code) *(Full-Stack)* — Fall 2025 – Spring 2026

- Built a full-stack TypeScript integration (calculus) contest platform with WebSocket matchmaking, JWT/Argon2 authentication, and role-based access control
- Developed competition tooling: LaTeX-rendered problem banks, configurable leaderboards, and a tag-based markdown wiki

COMPETITIONS & AWARDS

National Cyber League — Fall 2025

- Placed 2nd individually out of 1,000+ high school participants in a nationwide cybersecurity capture-the-flag (CTF) competition
- Placed 6th in the team round out of all high school teams

EXPERIENCE

Cybersecurity Club | Co-Captain — Fall 2024 – Spring 2026

- Led team strategy across forensics, cryptography, and web exploitation at four national CTF competitions; mentored members toward category specialization

Hackathon Club | President — Fall 2023 – Spring 2026

- Organized **LionHacks 2024 & 2025**, recruited judges, and grew event to **140+ participants**; coordinated cross-school logistics and finances for BISV Hacks 2025 & 2026
- Presented weekly technical sessions on topics such as JavaScript, React, and Firebase; mentored club members on full-stack web development

SKILLS

Cybersecurity: Ghidra, Pwntools, GDB, Metasploit, Wireshark, Burp Suite, Nmap

Software Development: Python, Java, C++, HTML/CSS/JavaScript, TypeScript, Bash, Rust

Infrastructure: Docker, Linux, Git, GitHub, Firebase, MongoDB

Data Science: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, Ultralytics YOLOv11, Jupyter Notebooks